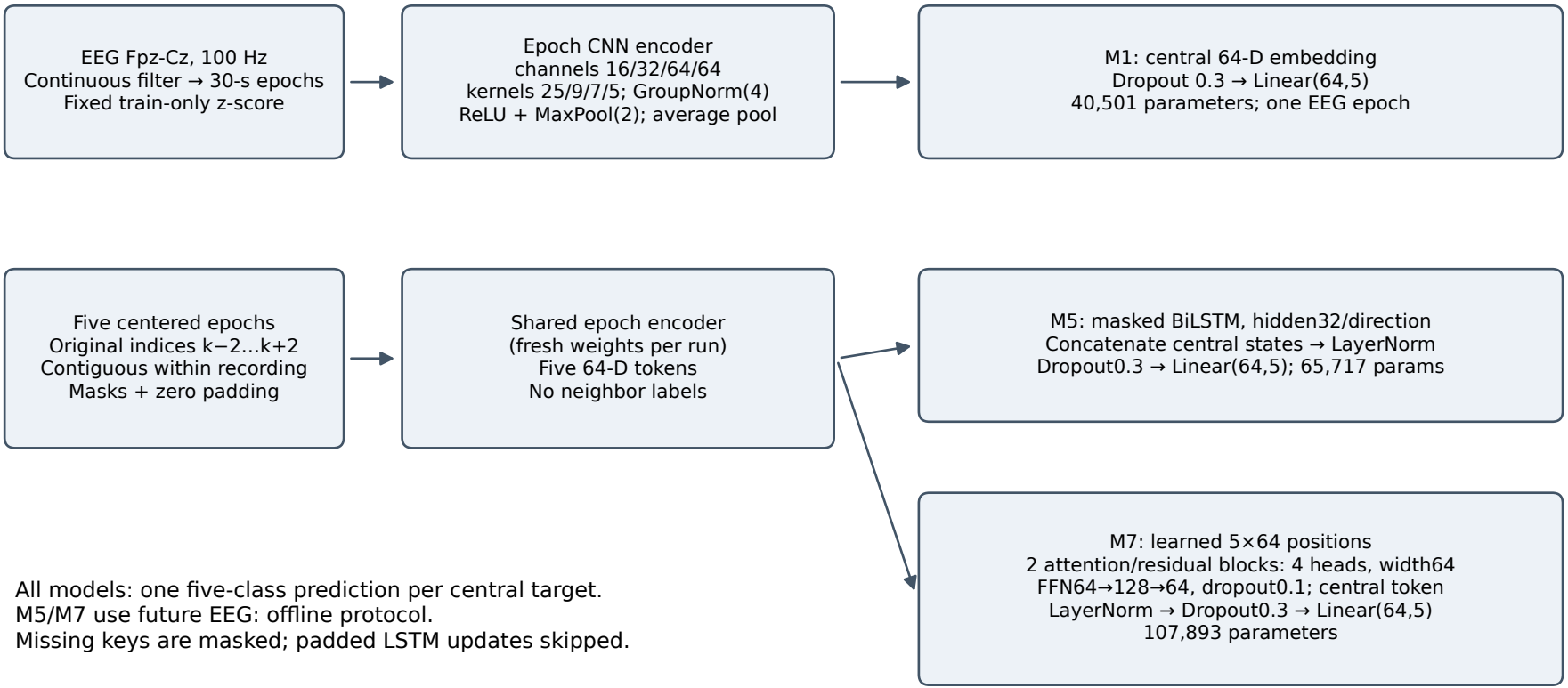
EEG Fpz-Cz, 100 Hz
Continuous filter → 30-s epochs
Fixed train-only z-score
Epoch CNN encoder
channels 16/32/64/64
kernels 25/9/7/5; GroupNorm(4)
ReLU + MaxPool(2); average pool
M1: central 64-D embedding
Dropout 0.3 → Linear(64,5)
40,501 parameters; one EEG epoch
Five centered epochs
Original indices k−2…k+2
Contiguous within recording
Masks + zero padding
Shared epoch encoder
(fresh weights per run)
Five 64-D tokens
No neighbor labels
M5: masked BiLSTM, hidden32/direction
Concatenate central states → LayerNorm
Dropout0.3 → Linear(64,5); 65,717 params
M7: learned 5×64 positions
2 attention/residual blocks: 4 heads, width64
FFN64→128→64, dropout0.1; central token
LayerNorm → Dropout0.3 → Linear(64,5)
107,893 parameters
All models: one five-class prediction per central target.
M5/M7 use future EEG: offline protocol.
Missing keys are masked; padded LSTM updates skipped.